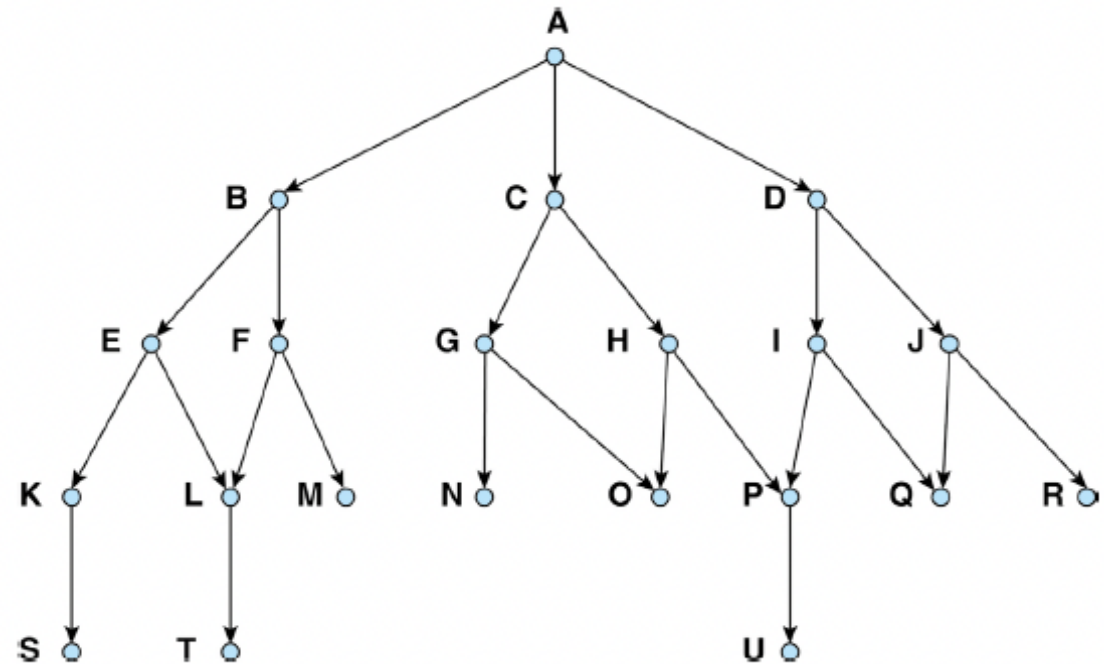


Activity

- Perform DFS and BFS state space search on the given tree.
- Suppose the state space is to be searched, starting at node "A" to goal "G".
- List the nodes of the state space in the order that they are expanded for each of the methods and the order that children of each expanded node are added to the queue.
- Assume that the search considers child nodes from left to right.



Algorithms

DFS

- begin
 - Open = [Start];
 - Closed = [];
 - While open <> [] do
 - begin
 - remove leftmost state from open, call it X;
 - if X is a goal then return(success)
 - else begin
 - generate children of X;
 - put X on closed;
 - discard children of X if already on open or closed;
 - put remaining children on left end of open
 - end
 - end
 - return(failure)
- end

BFS

- begin
 - Open = [Start];
 - Closed = [];
 - While open <> [] do
 - begin
 - remove leftmost state from open, call it X;
 - if X is a goal then return(success)
 - else begin
 - generate children of X;
 - put X on closed;
 - discard children of X if already on open or closed;
 - put remaining children on right end of open
 - end
 - end
 - return(failure)
- end